

SoC DESIGN – WEEK 6

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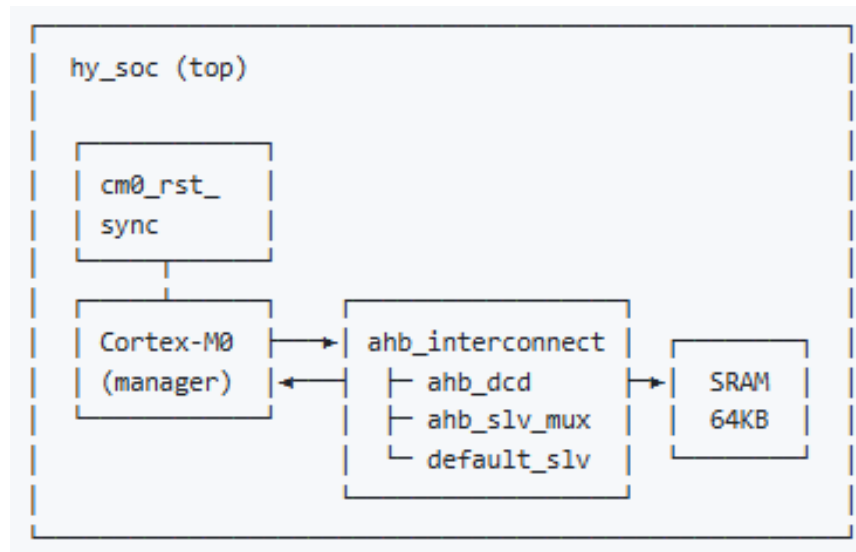
TA

Professor: Ji-Hoon Kim

Lab1

- Cortex-M0 + Default Slave

- Builds on Lab 0 by adding an AHB address decoder and Default Slave, so that accesses to unmapped addresses return an ERROR response.
 - Address Decoding (Address Phase): ahb_dcd examines HADDR to select the target subordinate
 - Response Multiplexing (Data Phase): ahb_slv_mux latches the select signal in the address phase and muxes the response in the data phase
 - Default Slave: Unmapped address → HRESP=1 (ERROR) → CM0 HardFault



Changes from Lab 0

Type	Description
New	<code>ahb_dcd.v</code> — Address decoder (SRAM vs Default)
New	<code>ahb_slv_mux.v</code> — Subordinate response multiplexer
New	<code>ahb_default_slv</code> — Returns ERROR response (2-cycle) for unmapped addresses
Modified	<code>ahb_interconnect.v</code> — Expanded to structural wrapper (DCD + MUX + Default Slave)

Directory Structure

- Cortex-M0 + Default Slave

Directory Structure

```
lab1_m0_default_slv/  
├── rtl/  
│   ├── hy_soc.v           ← Top-level module  
│   ├── ahb_interconnect.v ← Interconnect (DCD + MUX + Default Slave)  
│   ├── ahb_dcd.v          ← Address decoder  
│   └── ahb_slv_mux.v      ← Subordinate response mux  
├── tb/  
│   └── tb_hy_soc.v        ← Testbench  
├── sw/  
│   ├── test.c             ← Test firmware  
│   ├── gcc/               ← GCC build  
│   └── arm/               ← ARM Compiler build  
└── Makefile
```

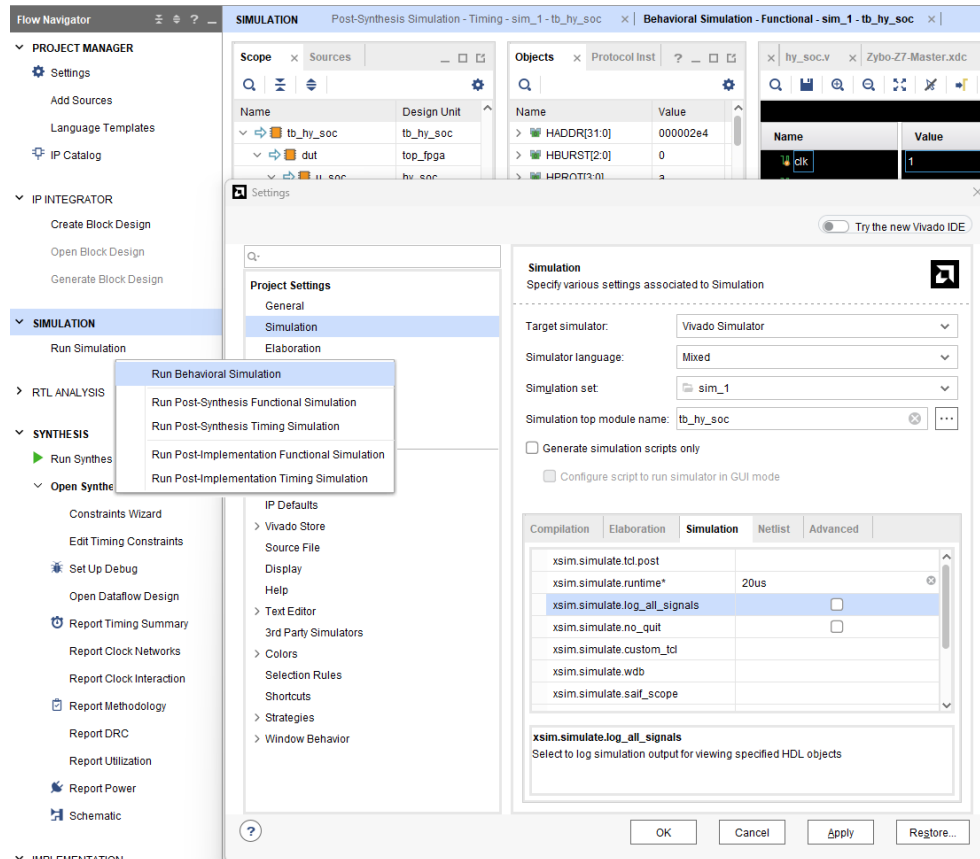
Memory Map

Address Range	Device	Notes
0x0000_0000 – 0x0000_FFFF	SRAM (64KB)	haddr[31:16] == 16'h0000
All other	Default Slave	ERROR response → HardFault

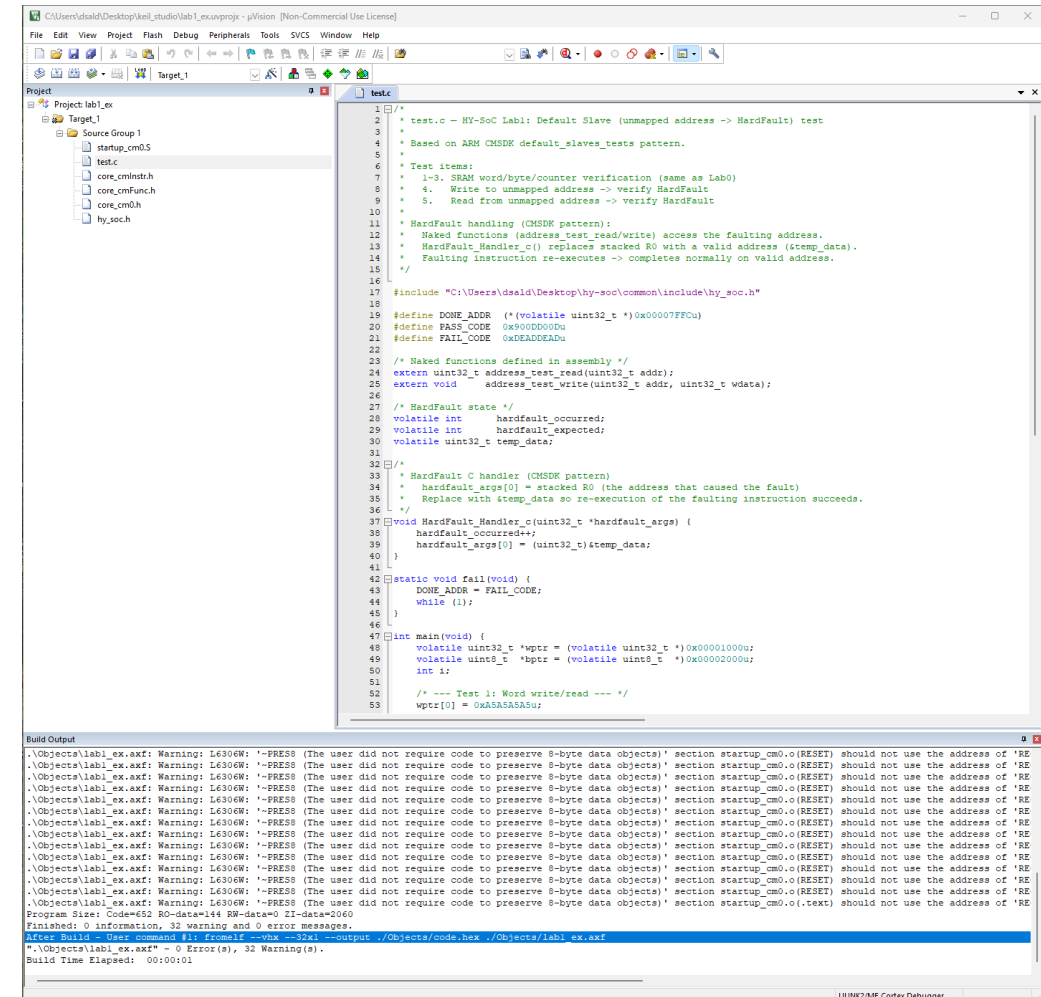
- Cortex-M0



• Run Simulation

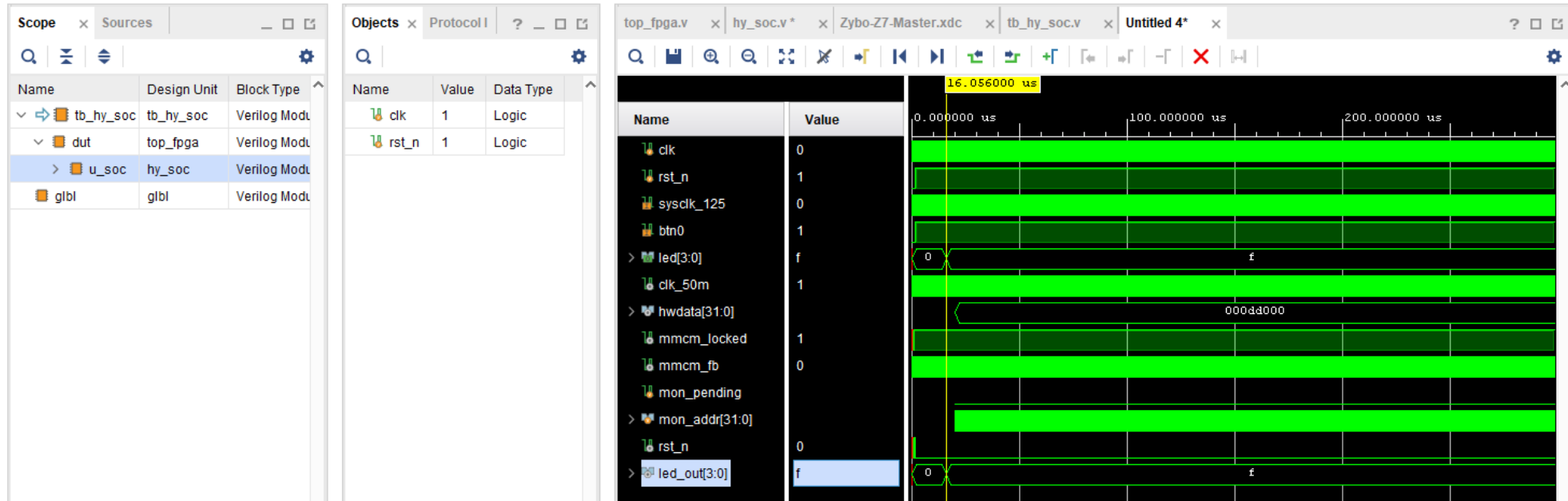


• Keil Screenshot (example. Test.c 보이게)



To-Do

- Lab1
 - Simulation with the pre-compiled code (Capture Full Screen)
 - Simulation Passed Code MDK Keil + test.c code (Capture Full Screen, Only one)
 - FPGA Board Demo
- FPGA

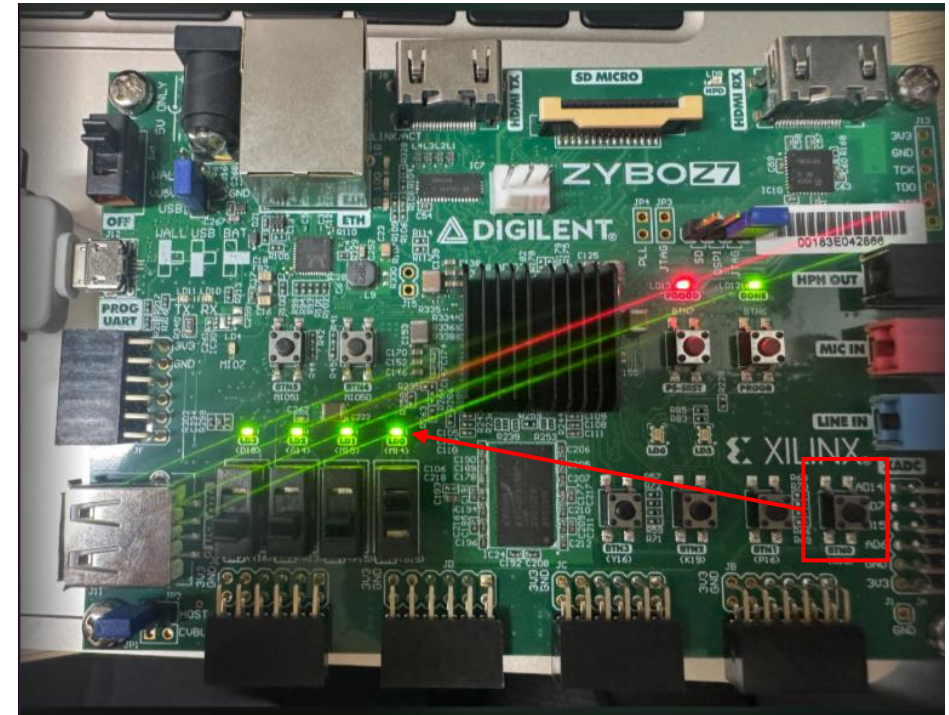
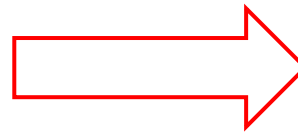


To-Do

- Keyword
 - {DONE_ADDR, PASS_CODE, FAIL_CODE, hwdata, mon_addr, mon_pending}

```
top_fpga.v x hy_soc.v x Zybo-Z7-Master.xdc x th_hy_soc.v x Untitled 4* x
C:\Users\dsald\Desktop\hy-soc\edutab1_m0_default_shiftrtlhy_soc.v

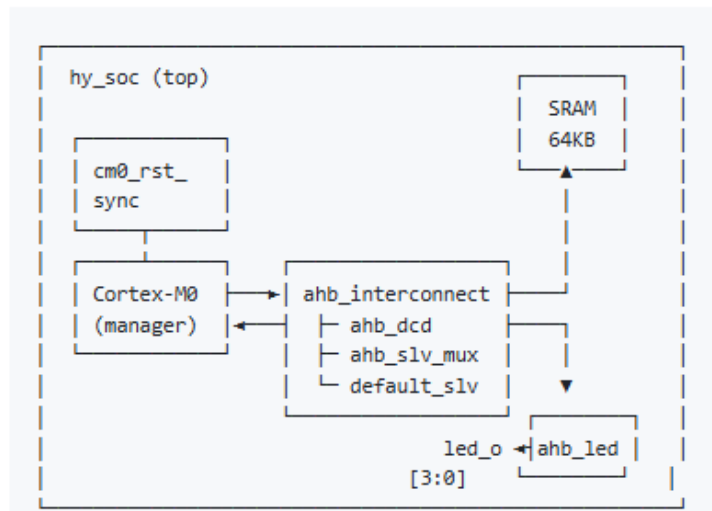
177 .hreadyout (hreadyout_s0),
178 .hdata (hdata_s0)
179 );
180
181 localparam DONE_ADDR = 32'h0007FFC;
182 localparam PASS_CODE = 32'h000D00D;
183 localparam FAIL_CODE = 32'hEADCEAD;
184
185 reg mon_pending;
186 reg [31:0] mon_addr;
187
188 always @(posedge clk or negedge rst_n) begin
189     if (!rst_n) begin
190         mon_pending <= 1'b0;
191         mon_addr <= 32'h0;
192     end else if (hready) begin
193         mon_pending <= hsel_s0 && hwrite && htrans[1];
194         mon_addr <= haddr;
195     end
196 end
197
198 // state
199 reg [3:0] c_led, n_led;
200
201 // sequential
202 always @(posedge clk or negedge rst_n) begin
203     if (!rst_n)
204         c_led <= 4'b0000;
205     else
206         c_led <= n_led;
207 end
208
209 // combinational
210 always @(*) begin
211     n_led = c_led; // default (82)
212
213     if ( [REDACTED] ) begin
214         if ( [REDACTED] )
215             n_led = 4'b1111;
216         else if ( [REDACTED] )
217             n_led = 4'b1010;
218     end
219 end
220
221 assign led = c_led;
222 endmodule
223
```



Lab2

- Cortex-M0 + LED Controller
 - Builds on Lab 1 by adding an LED controller peripheral, implementing SysTick interrupt-driven LED blinking. The AHB interconnect is expanded to handle two subordinates.
 - Peripheral Addition Pattern: Add address region to DCD → expand MUX → instantiate in top
 - SysTick Interrupt: CM0 built-in timer for periodic LED toggling
 - Simulation Mode: SIMULATION=1 macro shortens the blink period for faster simulation

Block Diagram



Changes from Lab 1

Type	Description
New	<code>ahb_led</code> — LED controller (4-bit output)
Modified	<code>hy_soc.v</code> — Added <code>led_o[3:0]</code> port and LED instance
Modified	<code>ahb_dcd.v</code> — Added LED address region (2-to-1 decoding)
Modified	<code>ahb_slv_mux.v</code> — Expanded to 3-way mux (SRAM + LED + Default)

Directory Structure

- Cortex-M0 + LED Controller

Directory Structure

```
lab2_m0_led/
├── rtl/
│   ├── hy_soc.v           ← Top-level (led_o port added)
│   ├── ahb_interconnect.v ← Interconnect (2 subordinates)
│   ├── ahb_dcd.v          ← Address decoder (SRAM + LED)
│   └── ahb_slv_mux.v       ← 3-way response mux
├── tb/
│   └── tb_hy_soc.v         ← Testbench (LED output monitoring)
├── sw/
│   ├── test.c             ← SysTick LED blink test
│   ├── gcc/               ← GCC build
│   └── arm/               ← ARM Compiler build
└── Makefile
```

Memory Map

Address Range	Device	Notes
0x0000_0000 – 0x0000_FFFF	SRAM (64KB)	haddr[31:16] == 16'h0000
0x5000_0000 – 0x50FF_FFFF	LED Controller	haddr[31:24] == 8'h50
All other	Default Slave	ERROR response

LED Registers

Offset	Name	Description
0x00	DATA	LED output [3:0] (R/W)

To-Do

- Lab2
 - Simulation Passed Code MDK Keil + test.c code (Capture Full Screen, Only one)
 - FPGA Board Demo
 - Provided Code : reset 이후, 4'b0000 -> 4'b1111 -> 4'b0000 -> 4'b1111 -> 4'b0000
 - To-Do Keil : reset 이후, Blink 횟수 늘리기 (10회)

THANK YOU

Any questions or comments:  Issu07@hanyang.ac.kr